UNIVERSITY OF ENGINEERING AND TECHNOLOGY

MARDAN

Name	**ASIM AYAZ**
Registration No#	**25MDBCS804**
Assignment	**Lab Assignment 5**
Submitted to	**Mam Alishba**
Semester	**2nd**
Date	**6-03-2026**

Q no 1:

Create a class as testing having integer variable as “a” and a static integer variable as “count”. The two member functions should be non-static and a static function of your own choice. Call nonstatic function thrice and it should increment the value of “count” and “a” private data member. The static function should display the private data member values upon calling from main function.

Source Code:

```cpp
#include <iostream>
using namespace std;
class Testing{
    private:
    int a=0;
    static int count;
    public:
    void  levelup() {
        a++;
          count++;
    }
    static void display(){
    cout<<"== We Are Going To Count the incremented a and count using static =="<<endl;
    cout<<"Count is "<<count<<endl;
    }
};
int Testing::count=0;
int main(){
    Testing t;
    t.levelup();
    t.levelup();
    t.levelup();

    t.display();
    return 0;
}
```

Output:

E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code\Lab5_2.exe

```
== We Are Going To Count the incremented a and count using static ==
Count is 3

--------------------------------
Process exited after 0.3544 seconds with return value 0
Press any key to continue . . .
```

Q no 2:

Create a class in C++ that counts the created objects using static member functions.

Source Code:

```cpp
#include <iostream>
using namespace std;

class Test {
private:
    static int objectCount;

public:
    Test() {
        objectCount++;
    }

    static void showCount() {
        cout << "Objects created = " << objectCount << endl;
    }
};

int Test::objectCount = 0;

int main() {
    Test a1, a2, a3,s2;

    Test::showCount();
}
```

Output:

```
E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code\Lab5_1.exe
Objects created = 4

--------------------------------
Process exited after 0.3435 seconds with return value 0
Press any key to continue . . .
```

Q no 3:

Write a C++ code that implement the concept of as how to declare, define and access a public static data member inside main function using class name.

```cpp
#include <iostream>
using namespace std;

class Sample {
public:
    static int value;// This is declaration
};

int Sample::value = 100;// via scope esolution

int main() {
    cout << Sample::value << endl;

    Sample::value = 200;// i update it

    cout << Sample::value << endl;
}
```

Output:

E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code\Lab5_3.exe

```
100
200

--------------------------------
Process exited after 0.4204 seconds with return value 0
Press any key to continue . . .
```